

Axiomatic System

Axiom 1 (Infinite Adjunction). Beginning with a base set P_0 , whenever a set P_n has been constructed, there exists a fixed distinguished element

$$\alpha_{n+1} \notin P_n$$

which has not appeared at any preceding construction stage. Distinct stages use distinct elements. Hence the construction may continue indefinitely.

At this point the subscripts

$$0, 1, 2, \dots$$

serve only as construction-stage labels. Their arithmetic properties have not yet been assumed or defined.

Axiom 2 (Function Well-Definedness). A function $f : X \rightarrow Y$ maps every input $x \in X$ to exactly one output $f(x) \in Y$. We denote evaluation by

$$f(x) \rightarrow y.$$

Axiom 3 (Structural Induction). Let $\mathcal{P}$ be a family generated from a base object P_0 by repeated application of a function S . If a property holds for P_0 , and whenever it holds for a constructed set P_n it also holds for $S(P_n)$, then the property holds for every set generated from P_0 by repeated applications of S .

Definitions and Rules

1.1 Base Set

Let

$$P_0 = \emptyset.$$

The empty set serves as the base representation of the symbol

$$0.$$

Thus

$$E(P_0) = 0.$$

1.2 Canonical Finite Sets

Using Axiom 1, define the canonical sets recursively.

Whenever P_n has already been constructed, define

$$P_{n+1} = P_n \cup \{\alpha_{n+1}\},$$

where

$$\alpha_{n+1} \notin P_n.$$

Consequently,

$$\begin{aligned} P_0 &= \emptyset, \\ P_1 &= \{\alpha_1\}, \\ P_2 &= \{\alpha_1, \alpha_2\}, \\ P_3 &= \{\alpha_1, \alpha_2, \alpha_3\}, \end{aligned}$$

and so on.

Let $\mathcal{P}$ denote the smallest family containing P_0 and closed under this construction.

Thus

$$\mathcal{P} = \{P_0, P_1, P_2, P_3, \dots\}.$$

Because the element introduced at each stage is fixed, each construction stage has a unique output.

1.3 Output Symbols and the Cardinality Function

We now assign one output symbol to each construction stage.

Define

$$E : \mathcal{P} \rightarrow \{0, 1, 2, 3, \dots\}$$

recursively by

$$E(P_0) = 0,$$

and whenever P_{n+1} is the immediate successor of P_n , let $E(P_{n+1})$ be the next output symbol after $E(P_n)$.

Accordingly,

$$\begin{aligned} E(P_0) &= 0, \\ E(P_1) &= 1, \\ E(P_2) &= 2, \\ E(P_3) &= 3, \end{aligned}$$

and so on.

Hence, after these symbols have been generated,

$$E(P_n) = n.$$

The statement $E(P_n) = n$ therefore does not use a previously defined arithmetic operation. It associates the symbol generated at stage n with the canonical set generated at the same stage.

Since each P_n contains exactly the elements introduced during the first n nonzero construction stages, $E(P_n)$ is also the finite cardinality of P_n . Therefore, once the symbols have been constructed, we may write

$$E(P_n) = |P_n|.$$

The domain of E in this construction is $\mathcal{P}$, so E is never required to assign a finite natural-number output to an arbitrary infinite set.

1.4 Successor Function

Define

$$S : \mathcal{P} \rightarrow \mathcal{P}$$

by

$$S(P_n) = P_n \cup \{\alpha_{n+1}\} = P_{n+1}.$$

Because α_{n+1} is fixed at construction stage $n + 1$, S maps every canonical set to a unique canonical set and is therefore well-defined.

Applying S repeatedly gives

$$\begin{aligned} P_0 &= \emptyset, \\ S(P_0) &= P_1 = \{\alpha_1\}, \\ S(S(P_0)) &= P_2 = \{\alpha_1, \alpha_2\}, \\ S(S(S(P_0))) &= P_3 = \{\alpha_1, \alpha_2, \alpha_3\}. \end{aligned}$$

Evaluating with E ,

$$\begin{aligned} E(P_0) &= 0, \\ E(S(P_0)) &= 1, \\ E(S(S(P_0))) &= 2, \\ E(S(S(S(P_0)))) &= 3. \end{aligned}$$

For any already constructed natural-number symbol k , write

$$S^k(P)$$

for k repeated applications of S , with

$$S^0(P) = P.$$

1.5 Natural Numbers Definition

The natural numbers are defined to be exactly the output symbols produced by evaluating E on the canonical sets:

$$\mathbb{N} = \{E(P) \mid P \in \mathcal{P}\}.$$

Thus

$$\boxed{\mathbb{N} = \{0, 1, 2, 3, \dots\}}.$$

This definition does not require a statement of the form $k \in \mathbb{N}$ before $\mathbb{N}$ itself has been constructed.

Defining Addition

We define addition using the canonical sets, the successor function S , and the evaluation function E .

For $P_1, P_2 \in \mathcal{P}$, define

$$\boxed{A(P_1, P_2) = E(S^{E(P_2)}(P_1))}.$$

Thus $A(P_1, P_2)$ is a natural-number symbol, not a set.

We define the ordinary notation $+$ by

$$\boxed{E(P_1) + E(P_2) := A(P_1, P_2)}.$$

Therefore,

$$E(P_1) + E(P_2) = E(S^{E(P_2)}(P_1)).$$

Addition by Zero

Since

$$E(P_0) = 0,$$

we have

$$A(P_1, P_0) = E(S^{E(P_0)}(P_1)) = E(S^0(P_1)) = E(P_1).$$

Hence

$$\boxed{E(P_1) + 0 = E(P_1)}.$$

Also,

$$A(P_0, P_1) = E(S^{E(P_1)}(P_0)).$$

By construction, applying S to P_0 exactly $E(P_1)$ times produces P_1 . Therefore

$$A(P_0, P_1) = E(P_1),$$

so

$$\boxed{0 + E(P_1) = E(P_1)}.$$

Successor Compatibility

Suppose

$$A(P_1, P_2) = E(P_r).$$

Then

$$S^{E(P_2)}(P_1) = P_r.$$

Since $S(P_2)$ contains exactly one more constructed element than P_2 ,

$$S^{E(S(P_2))}(P_1) = S(S^{E(P_2)}(P_1)).$$

Therefore

$$A(P_1, S(P_2)) = E(S(S^{E(P_2)}(P_1))) = E(S(P_r)).$$

Likewise,

$$S^{E(P_1)}(S(P_2)) = S(S^{E(P_1)}(P_2)).$$

Thus taking one successor step in either input produces one successor step in the resulting canonical set.

Commutativity of Addition

Theorem 1 *For all $P_1, P_2 \in \mathcal{P}$,*

$$A(P_1, P_2) = A(P_2, P_1).$$

Consequently,

$$E(P_1) + E(P_2) = E(P_2) + E(P_1).$$

Fix P_1 . We use structural induction on P_2 .

For the base case,

$$P_2 = P_0.$$

Then

$$A(P_1, P_0) = E(P_1).$$

Also,

$$A(P_0, P_1) = E(S^{E(P_1)}(P_0)) = E(P_1).$$

Hence

$$A(P_1, P_0) = A(P_0, P_1).$$

Now assume

$$A(P_1, P_2) = A(P_2, P_1).$$

Let

$$P'_2 = S(P_2).$$

By successor compatibility,

$$A(P_1, P'_2)$$

is the output symbol immediately following

$$A(P_1, P_2).$$

Likewise,

$$A(P'_2, P_1)$$

is the output symbol immediately following

$$A(P_2, P_1).$$

By the induction hypothesis,

$$A(P_1, P_2) = A(P_2, P_1),$$

so their immediate successor symbols are equal. Therefore

$$A(P_1, P'_2) = A(P'_2, P_1).$$

By structural induction,

$$A(P_1, P_2) = A(P_2, P_1)$$

for every $P_2 \in \mathcal{P}$.

Therefore it is valid to write

$$\boxed{E(P_1) + E(P_2) = E(P_2) + E(P_1)}.$$

Example: $1 + 2$

Let

$$P_0 = \emptyset,$$

$$P_1 = S(P_0),$$

$$P_2 = S(S(P_0)).$$

Then

$$E(P_1) = 1, \quad E(P_2) = 2.$$

Evaluate

$$A(P_1, P_2) = E \left(S^{E(P_2)}(P_1) \right).$$

Since

$$E(P_2) = 2,$$

we obtain

$$A(P_1, P_2) = E(S^2(P_1)).$$

But

$$S^2(P_1) = S(S(P_1)) = P_3.$$

Thus

$$A(P_1, P_2) = E(P_3) = 3.$$

Therefore

$$1 + 2 = 3.$$

By the commutativity theorem,

$$A(P_2, P_1) = A(P_1, P_2) = 3,$$

so

$$2 + 1 = 3.$$

Hence

$$\boxed{1 + 2 = 2 + 1 = 3.}$$

Subtraction on Natural Numbers via Element Removal

We now define subtraction when the result remains nonnegative.

1. Element Removal Function

Define the removal function N on every nonempty canonical set by

$$\boxed{N(P_n) = P_{n-1} \quad \text{for } P_n \neq P_0.}$$

Equivalently, N removes the distinguished element introduced most recently in the canonical construction.

For example,

$$P_3 = \{\alpha_1, \alpha_2, \alpha_3\}.$$

Then

$$N(P_3) = \{\alpha_1, \alpha_2\} = P_2.$$

Similarly,

$$N(P_2) = P_1$$

and

$$N(P_1) = P_0.$$

In particular,

$$N(P_1) = P_0 = \emptyset,$$

and therefore

$$E(N(P_1)) = 0.$$

The rule specifies a unique element to remove, so N is well-defined.

2. Iterated Removal

For any natural-number symbol k , define

$$N^0(P) = P,$$

and, whenever the intermediate set is nonempty,

$$N^k(P) = N(N^{k-1}(P))$$

for nonzero k .

For example,

$$N^2(P_3) = N(N(P_3)) = N(P_2) = P_1.$$

3. Ordering of Natural Numbers

Before restricting subtraction, define the order relation $\leq$.

For canonical sets P_1, P_2 , define

$$\boxed{E(P_2) \leq E(P_1)}$$

to mean that P_1 can be obtained from P_2 by zero or more applications of S .

Equivalently,

$$E(P_2) \leq E(P_1)$$

when there exists a natural-number symbol k such that

$$S^k(P_2) = P_1.$$

Thus, for example,

$$1 \leq 3$$

because

$$S^2(P_1) = P_3.$$

4. Definition of Natural Subtraction

Let

$$D(P_1, P_2)$$

be defined whenever

$$E(P_2) \leq E(P_1).$$

Define

$$\boxed{D(P_1, P_2) = E(N^{E(P_2)}(P_1))}.$$

We then define

$$\boxed{E(P_1) - E(P_2) := D(P_1, P_2)}$$

whenever

$$E(P_2) \leq E(P_1).$$

The restriction guarantees that N is never required to remove an element from P_0 .

5. Example Evaluation

Let

$$P_3 = \{\alpha_1, \alpha_2, \alpha_3\}$$

and

$$P_1 = \{\alpha_1\}.$$

Then

$$E(P_3) = 3$$

and

$$E(P_1) = 1.$$

Evaluate

$$D(P_3, P_1) = E(N^{E(P_1)}(P_3)).$$

Since

$$E(P_1) = 1,$$

we have

$$D(P_3, P_1) = E(N(P_3)).$$

By the definition of N ,

$$N(P_3) = P_2.$$

Therefore

$$D(P_3, P_1) = E(P_2) = 2.$$

Thus

$$\boxed{3 - 1 = 2}.$$

Constructing Negative Numbers

Natural-number subtraction as defined above does not yet assign an output to expressions such as

$$1 - 2.$$

We now construct signed integers using ordered pairs of canonical sets.

1. Ordered-Pair Representation

Take an ordered pair

$$(P_x, P_y),$$

where

$$P_x, P_y \in \mathcal{P}.$$

The first component records a positive contribution, and the second component records a negative contribution.

At this stage, however, the symbol “negative” describes the role of the second component only. A negative-number symbol such as -1 will be introduced only after the corresponding ordered-pair object has been constructed.

2. Reduction Function

Define

$$R[(P_x, P_y)]$$

recursively by

$$R[(P_x, P_y)] = \begin{cases} R[(N(P_x), N(P_y))], & E(P_x) > 0 \text{ and } E(P_y) > 0, \\ (P_x, P_y), & E(P_x) = 0 \text{ or } E(P_y) = 0. \end{cases}$$

Here

$$E(P) > 0$$

means simply

$$P \neq P_0.$$

Every recursive step removes one element from both components.

Since P_x and P_y are finite canonical sets, after finitely many applications of N , at least one component must become P_0 .

Therefore R always terminates.

Most importantly,

$$R$$

always returns an ordered pair. Its output type does not change during the definition.

3. Reduced Signed Pairs

A pair

$$(P_x, P_y)$$

is called *reduced* if at least one component equals P_0 .

Thus every reduced pair has exactly one of the following forms:

$$(P_n, P_0),$$

$$(P_0, P_n),$$

or

$$(P_0, P_0).$$

The function R maps every ordered pair of canonical sets to exactly one reduced pair.

4. Signed Integer Notation

We now introduce notation for these already-constructed reduced pairs.

For a nonzero natural-number symbol n , define the notation

$$(P_n, P_0) \equiv n.$$

Define

$$(P_0, P_0) \equiv 0.$$

Finally, for a nonzero natural-number symbol n , introduce the new notation

$$\boxed{(P_0, P_n) \equiv -n.}$$

Thus the symbol

$$-n$$

is not used to construct the negative integer.

Instead, the ordered pair

$$(P_0, P_n)$$

is constructed first, and $-n$ is then assigned as notation for that object.

The signed integers are therefore the reduced pairs

$$\boxed{\mathbb{Z} = \{R[(P_x, P_y)] \mid P_x, P_y \in \mathcal{P}\}.$$

Using the notation above, these objects are written

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

5. Canonical Representation

Because R removes one element from each side simultaneously, every pair has a unique reduced representative.

For example,

$$R[(P_2, P_1)] = R[(N(P_2), N(P_1))] = R[(P_1, P_0)] = (P_1, P_0).$$

Thus

$$(P_2, P_1)$$

and

$$(P_1, P_0)$$

reduce to the same signed integer.

More generally, two ordered pairs

$$(P_a, P_b)$$

and

$$(P_c, P_d)$$

represent the same signed integer precisely when

$$\boxed{R[(P_a, P_b)] = R[(P_c, P_d)].}$$

Hence signed integers have unique reduced representatives even though unreduced intermediate pairs may differ.

6. Examples

Example 1: $1 - 2$.

Take

$$P_1 = \{\alpha_1\}$$

and

$$P_2 = \{\alpha_1, \alpha_2\}.$$

Evaluate

$$R[(P_1, P_2)].$$

Since both components are nonempty,

$$R[(P_1, P_2)] = R[(N(P_1), N(P_2))].$$

Using

$$N(P_1) = P_0$$

and

$$N(P_2) = P_1,$$

we obtain

$$R[(P_1, P_2)] = R[(P_0, P_1)].$$

Because the first component is now P_0 ,

$$R[(P_0, P_1)] = (P_0, P_1).$$

By the signed notation introduced above,

$$(P_0, P_1) \equiv -1.$$

Therefore

$$\boxed{1 - 2 = -1.}$$

Example 2: $2 - 1$.

Evaluate

$$R[(P_2, P_1)].$$

Since both components are nonempty,

$$R[(P_2, P_1)] = R[(N(P_2), N(P_1))].$$

Thus

$$R[(P_2, P_1)] = R[(P_1, P_0)] = (P_1, P_0).$$

By signed notation,

$$(P_1, P_0) \equiv 1.$$

Therefore

$$\boxed{2 - 1 = 1.}$$

Example 3: $2 - 2$.

Evaluate

$$R[(P_2, P_2)].$$

First,

$$R[(P_2, P_2)] = R[(N(P_2), N(P_2))] = R[(P_1, P_1)].$$

Both components remain nonempty, so

$$R[(P_1, P_1)] = R[(N(P_1), N(P_1))] = R[(P_0, P_0)].$$

The pair is now reduced, giving

$$R[(P_0, P_0)] = (P_0, P_0).$$

Since

$$(P_0, P_0) \equiv 0,$$

we obtain

$$\boxed{2 - 2 = 0.}$$

Addition and Subtraction of Signed Pairs

We now define arithmetic directly on reduced signed-pair representations.

Let

$$(P_a, P_b)$$

and

$$(P_c, P_d)$$

be signed-pair representations.

The positive contributions are represented by

$$P_a \quad \text{and} \quad P_c,$$

while the negative contributions are represented by

$$P_b \quad \text{and} \quad P_d.$$

Signed Addition

The combined positive count is

$$A(P_a, P_c).$$

The corresponding canonical set is

$$S^{A(P_a, P_c)}(P_0).$$

Likewise, the combined negative count is

$$A(P_b, P_d),$$

with corresponding canonical set

$$S^{A(P_b, P_d)}(P_0).$$

Therefore define

$$\text{Add}((P_a, P_b), (P_c, P_d)) = R \left[(S^{A(P_a, P_c)}(P_0), S^{A(P_b, P_d)}(P_0)) \right].$$

The result is a reduced signed pair.

Signed Additive Inverse

For a signed pair

$$(P_c, P_d),$$

define its additive inverse by exchanging its two components:

$$-(P_c, P_d) := (P_d, P_c).$$

This is valid because the component that formerly contributed positively now contributes negatively, and the component that formerly contributed negatively now contributes positively.

For example,

$$-(P_2, P_0) = (P_0, P_2),$$

so the additive inverse of 2 is represented by

$$(P_0, P_2) \equiv -2.$$

Similarly,

$$-(P_0, P_2) = (P_2, P_0),$$

so the additive inverse of -2 is 2.

Signed Subtraction

Subtraction is addition of the additive inverse.

Thus

$$(P_a, P_b) - (P_c, P_d)$$

is defined as

$$(P_a, P_b) + (P_d, P_c).$$

Therefore

$$\text{Subtract}((P_a, P_b), (P_c, P_d)) = R \left[(S^{A(P_a, P_d)}(P_0), S^{A(P_b, P_c)}(P_0)) \right].$$

1. Adding Same Signs: $(-2) + (-1)$

The inputs are

$$(P_0, P_2)$$

and

$$(P_0, P_1).$$

The new positive count is

$$A(P_0, P_0) = 0.$$

Therefore

$$P_{\text{pos}} = S^0(P_0) = P_0.$$

The new negative count is

$$A(P_2, P_1) = 3.$$

Therefore

$$P_{\text{neg}} = S^3(P_0) = P_3.$$

Hence

$$\text{Add}((P_0, P_2), (P_0, P_1)) = R[(P_0, P_3)].$$

Since the first component is already P_0 ,

$$R[(P_0, P_3)] = (P_0, P_3).$$

By signed notation,

$$(P_0, P_3) \equiv -3.$$

Therefore

$$\boxed{(-2) + (-1) = -3.}$$

2. Adding Opposite Signs: $(-2) + 1$

The inputs are

$$(P_0, P_2)$$

and

$$(P_1, P_0).$$

The new positive count is

$$A(P_0, P_1) = 1,$$

so

$$P_{\text{pos}} = P_1.$$

The new negative count is

$$A(P_2, P_0) = 2,$$

so

$$P_{\text{neg}} = P_2.$$

Thus

$$\text{Add}((P_0, P_2), (P_1, P_0)) = R[(P_1, P_2)].$$

Applying N once to both components gives

$$R[(P_1, P_2)] = R[(P_0, P_1)] = (P_0, P_1).$$

Since

$$(P_0, P_1) \equiv -1,$$

we obtain

$$\boxed{(-2) + 1 = -1.}$$

3. Subtracting Same Signs: $1 - 2$

The inputs are

$$(P_1, P_0)$$

and

$$(P_2, P_0).$$

Using the subtraction definition,

$$A(P_1, P_0) = 1,$$

so

$$P_{\text{pos}} = P_1.$$

Also,

$$A(P_0, P_2) = 2,$$

so

$$P_{\text{neg}} = P_2.$$

Thus

$$\text{Subtract}((P_1, P_0), (P_2, P_0)) = R[(P_1, P_2)].$$

Reduction gives

$$R[(P_1, P_2)] = R[(P_0, P_1)] = (P_0, P_1).$$

Since

$$(P_0, P_1) \equiv -1,$$

we obtain

$$\boxed{1 - 2 = -1.}$$

4. Subtracting Opposite Signs: $1 - (-2)$

The inputs are

$$(P_1, P_0)$$

and

$$(P_0, P_2).$$

The positive count is

$$A(P_1, P_2) = 3,$$

so

$$P_{\text{pos}} = P_3.$$

The negative count is

$$A(P_0, P_0) = 0,$$

so

$$P_{\text{neg}} = P_0.$$

Therefore

$$\text{Subtract } ((P_1, P_0), (P_0, P_2)) = R[(P_3, P_0)].$$

Since the second component is already P_0 ,

$$R[(P_3, P_0)] = (P_3, P_0).$$

By signed notation,

$$(P_3, P_0) \equiv 3.$$

Hence

$$\boxed{1 - (-2) = 3.}$$

Proof by Fully Expanded Substitution

We now expand the preceding signed arithmetic computations back to the base set P_0 using only

$$S(P), \quad E(P), \quad A(P_1, P_2), \quad N(P), \quad R[(P_x, P_y)].$$

1. Adding Same Signs: $(-2) + (-1) = -3$

The base representations are

$$\text{Pair 1} = (P_0, S(S(P_0)))$$

and

$$\text{Pair 2} = (P_0, S(P_0)).$$

The expanded addition formula is

$$\text{Add} = R \left[(S^{A(P_0, P_0)}(P_0), S^{A(S(S(P_0)), S(P_0))}(P_0)) \right].$$

First evaluate the positive component:

$$A(P_0, P_0) = E(S^{E(P_0)}(P_0)).$$

Since

$$E(P_0) = 0,$$

we obtain

$$A(P_0, P_0) = E(S^0(P_0)) = E(P_0) = 0.$$

Thus

$$k_{\text{pos}} = 0.$$

Now evaluate the negative component:

$$A(S(S(P_0)), S(P_0)) = E(S^{E(S(P_0))}(S(S(P_0)))).$$

Since

$$E(S(P_0)) = 1,$$

we obtain

$$A(S(S(P_0)), S(P_0)) = E(S^1(S(S(P_0)))).$$

Therefore

$$A(S(S(P_0)), S(P_0)) = E(S(S(S(P_0)))) = 3.$$

Thus

$$k_{\text{neg}} = 3.$$

Construct the new sets:

$$P_x = S^0(P_0) = P_0,$$

and

$$P_y = S^3(P_0) = S(S(S(P_0))).$$

Therefore

$$\text{Add} = R[(P_0, S(S(S(P_0))))].$$

Since the first component is P_0 , reduction stops immediately:

$$R[(P_0, S(S(S(P_0))))] = (P_0, S(S(S(P_0)))).$$

But

$$S(S(S(P_0))) = P_3.$$

Hence the reduced pair is

$$(P_0, P_3).$$

By signed notation,

$$(P_0, P_3) \equiv -3.$$

Therefore

$$\boxed{(-2) + (-1) = -3.}$$

2. Adding Opposite Signs: $(-2) + 1 = -1$

The base representations are

$$\text{Pair 1} = (P_0, S(S(P_0)))$$

and

$$\text{Pair 2} = (S(P_0), P_0).$$

The expanded addition formula is

$$\text{Add} = R \left[(S^{A(P_0, S(P_0))}(P_0), S^{A(S(S(P_0)), P_0)}(P_0)) \right].$$

Evaluate the positive component:

$$A(P_0, S(P_0)) = E(S^{E(S(P_0))}(P_0)).$$

Since

$$E(S(P_0)) = 1,$$

we have

$$A(P_0, S(P_0)) = E(S^1(P_0)) = E(S(P_0)) = 1.$$

Thus

$$k_{\text{pos}} = 1.$$

Now evaluate the negative component:

$$A(S(S(P_0)), P_0) = E(S^{E(P_0)}(S(S(P_0)))) .$$

Since

$$E(P_0) = 0,$$

we obtain

$$A(S(S(P_0)), P_0) = E(S^0(S(S(P_0)))) = E(S(S(P_0))) = 2.$$

Thus

$$k_{\text{neg}} = 2.$$

Construct the sets:

$$P_x = S^1(P_0) = S(P_0),$$

and

$$P_y = S^2(P_0) = S(S(P_0)).$$

Therefore

$$\text{Add} = R[(S(P_0), S(S(P_0)))].$$

Because both sets are nonempty,

$$R[(S(P_0), S(S(P_0)))] = R[(N(S(P_0)), N(S(S(P_0))))].$$

By the definition of N ,

$$N(S(P_0)) = P_0$$

and

$$N(S(S(P_0))) = S(P_0).$$

Therefore

$$R[(S(P_0), S(S(P_0)))] = R[(P_0, S(P_0))].$$

The first component is now P_0 , so

$$R[(P_0, S(P_0))] = (P_0, S(P_0)).$$

Since

$$S(P_0) = P_1,$$

the reduced pair is

$$(P_0, P_1).$$

By signed notation,

$$(P_0, P_1) \equiv -1.$$

Therefore

$$\boxed{(-2) + 1 = -1.}$$

3. Subtracting Same Signs: $1 - 2 = -1$

The base representations are

$$\text{Pair 1} = (S(P_0), P_0)$$

and

$$\text{Pair 2} = (S(S(P_0)), P_0).$$

The expanded subtraction formula is

$$\text{Subtract} = R \left[(S^{A(S(P_0), P_0)}(P_0), S^{A(P_0, S(S(P_0)))}(P_0)) \right].$$

Evaluate the positive component:

$$A(S(P_0), P_0) = E(S^{E(P_0)}(S(P_0))) .$$

Since

$$E(P_0) = 0,$$

we obtain

$$A(S(P_0), P_0) = E(S^0(S(P_0))) = E(S(P_0)) = 1.$$

Thus

$$k_{\text{pos}} = 1.$$

Evaluate the negative component:

$$A(P_0, S(S(P_0))) = E(S^{E(S(P_0))}(P_0)) .$$

Since

$$E(S(S(P_0))) = 2,$$

we obtain

$$A(P_0, S(S(P_0))) = E(S^2(P_0)) = E(S(S(P_0))) = 2.$$

Thus

$$k_{\text{neg}} = 2.$$

Construct the sets:

$$P_x = S^1(P_0) = S(P_0),$$

and

$$P_y = S^2(P_0) = S(S(P_0)).$$

Therefore

$$\text{Subtract} = R[(S(P_0), S(S(P_0)))].$$

Reduction gives

$$R[(S(P_0), S(S(P_0)))] = R[(P_0, S(P_0))] = (P_0, S(P_0)).$$

Since

$$S(P_0) = P_1,$$

we obtain

$$(P_0, S(P_0)) = (P_0, P_1).$$

By signed notation,

$$(P_0, P_1) \equiv -1.$$

Therefore

$$\boxed{1 - 2 = -1.}$$

4. Subtracting Opposite Signs: $1 - (-2) = 3$

The base representations are

$$\text{Pair 1} = (S(P_0), P_0)$$

and

$$\text{Pair 2} = (P_0, S(S(P_0))).$$

The expanded subtraction formula is

$$\text{Subtract} = R \left[(S^{A(S(P_0), S(S(P_0)))}(P_0), S^{A(P_0, P_0)}(P_0)) \right].$$

Evaluate the positive component:

$$A(S(P_0), S(S(P_0))) = E(S^{E(S(S(P_0)))}(S(P_0))).$$

Since

$$E(S(S(P_0))) = 2,$$

we obtain

$$A(S(P_0), S(S(P_0))) = E(S^2(S(P_0))).$$

Therefore

$$A(S(P_0), S(S(P_0))) = E(S(S(S(P_0)))) = 3.$$

Thus

$$k_{\text{pos}} = 3.$$

Evaluate the negative component:

$$A(P_0, P_0) = E(S^{E(P_0)}(P_0)).$$

Since

$$E(P_0) = 0,$$

we obtain

$$A(P_0, P_0) = E(P_0) = 0.$$

Thus

$$k_{\text{neg}} = 0.$$

Construct the sets:

$$P_x = S^3(P_0) = S(S(S(P_0))),$$

and

$$P_y = S^0(P_0) = P_0.$$

Therefore

$$\text{Subtract} = R[(S(S(S(P_0))), P_0)].$$

Since the second component is already P_0 ,

$$R[(S(S(S(P_0))), P_0)] = (S(S(S(P_0))), P_0).$$

Since

$$S(S(S(P_0))) = P_3,$$

the reduced pair is

$$(P_3, P_0).$$

By signed notation,

$$(P_3, P_0) \equiv 3.$$

Therefore

$$\boxed{1 - (-2) = 3.}$$

Conclusion

Beginning with

$$P_0 = \emptyset,$$

the canonical sets are generated by the well-defined successor function

$$S(P_n) = P_{n+1}.$$

The evaluation function E associates each canonical construction stage with exactly one natural-number symbol:

$$E(P_n) = n.$$

This produces

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}.$$

Addition is then defined by

$$\boxed{A(P_1, P_2) = E\left(S^{E(P_2)}(P_1)\right),}$$

with

$$E(P_1) + E(P_2) := A(P_1, P_2).$$

Commutativity follows by structural induction:

$$\boxed{E(P_1) + E(P_2) = E(P_2) + E(P_1).}$$

For nonnegative results, subtraction is defined by the deterministic removal function

$$N(P_n) = P_{n-1}$$

and

$$D(P_1, P_2) = E(N^{E(P_2)}(P_1)).$$

Signed integers are then constructed as reduced ordered pairs of canonical sets. The reduction function

$$R[(P_x, P_y)] = \begin{cases} R[(N(P_x), N(P_y))], & E(P_x) > 0 \text{ and } E(P_y) > 0, \\ (P_x, P_y), & E(P_x) = 0 \text{ or } E(P_y) = 0 \end{cases}$$

always returns a reduced ordered pair.

Only after these ordered-pair objects have been constructed is signed notation introduced:

$$(P_n, P_0) \equiv n,$$

$$(P_0, P_0) \equiv 0,$$

and

$$(P_0, P_n) \equiv -n.$$

Signed addition is defined by

$$\text{Add}((P_a, P_b), (P_c, P_d)) = R[(S^{A(P_a, P_c)}(P_0), S^{A(P_b, P_d)}(P_0))],$$

while signed subtraction is defined by

$$\text{Subtract}((P_a, P_b), (P_c, P_d)) = R[(S^{A(P_a, P_d)}(P_0), S^{A(P_b, P_c)}(P_0))].$$

The explicit evaluations establish

$$1 + 2 = 3,$$

$$3 - 1 = 2,$$

$$1 - 2 = -1,$$

$$2 - 1 = 1,$$

$$2 - 2 = 0,$$

$$(-2) + (-1) = -3,$$

$$(-2) + 1 = -1,$$

and

$$1 - (-2) = 3.$$

Thus the natural-number and signed-integer operations used above are obtained from the base set P_0 , deterministic adjunction and removal, evaluation of canonical finite sets, and reduction of ordered pairs.

Non-Unique Signed-Pair Representations

The signed-pair construction admits infinitely many unreduced representations of the same signed integer.

Let

$$(P_a, P_b)$$

be any signed-pair representation, and let

$$P_v \in \mathcal{P}.$$

Construct the shifted pair

$$(S^{E(P_v)}(P_a), S^{E(P_v)}(P_b)).$$

Since

$$E(P_v) = v,$$

and applying S exactly v times to a canonical set P_a produces P_{a+v} , we have

$$S^{E(P_v)}(P_a) = P_{a+v}$$

and

$$S^{E(P_v)}(P_b) = P_{b+v}.$$

Hence the shifted pair may be written

$$\boxed{(S^{E(P_v)}(P_a), S^{E(P_v)}(P_b)) = (P_{a+v}, P_{b+v}).}$$

Invariance Under Equal Shifting

Theorem 2 *For every $P_a, P_b, P_v \in \mathcal{P}$,*

$$\boxed{R[(P_a, P_b)] = R[(P_{a+v}, P_{b+v})].}$$

Equivalently,

$$\boxed{R[(P_a, P_b)] = R[(S^{E(P_v)}(P_a), S^{E(P_v)}(P_b))].}$$

The pair

$$(P_{a+v}, P_{b+v})$$

contains the same additional contribution v in both components.

By the definition of R , whenever both components are nonempty, one application of N is made to each component simultaneously.

Applying this simultaneous removal v times gives

$$R[(P_{a+v}, P_{b+v})] = R[(P_a, P_b)].$$

Therefore adding the same canonical contribution to both components does not change the reduced signed integer represented by the pair.

Hence

$$\boxed{R[(P_{a+v}, P_{b+v})] = R[(P_a, P_b)]}.$$

Thus a single signed integer does not have only one possible unreduced representation.

For example,

$$(P_a, P_b),$$

$$(P_{a+1}, P_{b+1}),$$

$$(P_{a+2}, P_{b+2}),$$

and, more generally,

$$(P_{a+v}, P_{b+v})$$

all reduce to the same signed integer.

Because the canonical family

$$\mathcal{P} = \{P_0, P_1, P_2, P_3, \dots\}$$

contains infinitely many canonical sets, there are infinitely many possible choices of P_v , and therefore infinitely many unreduced representations of the same signed integer.

Non-Unique Input Representations for Signed Addition

Consider

$$\text{Add}((P_a, P_b), (P_c, P_d)).$$

Let

$$P_v, P_w \in \mathcal{P}.$$

Independently shift the two input pairs by P_v and P_w , respectively.

The first pair becomes

$$(S^{E(P_v)}(P_a), S^{E(P_v)}(P_b)),$$

and the second becomes

$$(S^{E(P_w)}(P_c), S^{E(P_w)}(P_d)).$$

Therefore,

$$\begin{aligned} & \text{Add}((P_a, P_b), (P_c, P_d)) \\ &= \text{Add}((S^{E(P_v)}(P_a), S^{E(P_v)}(P_b)), (S^{E(P_w)}(P_c), S^{E(P_w)}(P_d))). \end{aligned}$$

Using the canonical-set notation, this is equivalently

$$\text{Add}((P_a, P_b), (P_c, P_d)) = \text{Add}((P_{a+v}, P_{b+v}), (P_{c+w}, P_{d+w})).$$

Theorem 3 *For every*

$$P_a, P_b, P_c, P_d, P_v, P_w \in \mathcal{P},$$

the equality

$$\text{Add}((P_a, P_b), (P_c, P_d)) = \text{Add}((P_{a+v}, P_{b+v}), (P_{c+w}, P_{d+w}))$$

holds.

By the preceding theorem,

$$R[(P_{a+v}, P_{b+v})] = R[(P_a, P_b)]$$

and independently,

$$R[(P_{c+w}, P_{d+w})] = R[(P_c, P_d)].$$

Thus each shifted pair represents exactly the same signed integer as its corresponding original pair.

The definition of signed addition combines the positive components, combines the negative components, and then applies R .

Adding v to both components of the first pair contributes the same additional amount to the positive and negative sides.

Likewise, adding w to both components of the second pair contributes the same additional amount to the positive and negative sides.

These equal contributions cancel under R .

Therefore

$$\text{Add}((P_{a+v}, P_{b+v}), (P_{c+w}, P_{d+w})) = \text{Add}((P_a, P_b), (P_c, P_d)).$$

Infinitely Many Addition Representations

Since P_v and P_w may each be chosen independently from the infinite family

$$\mathcal{P},$$

a single signed addition may be represented by infinitely many different input tuples.

Thus

$$\text{Add}((P_a, P_b), (P_c, P_d))$$

may equivalently be evaluated using

$$\text{Add}((P_{a+v}, P_{b+v}), (P_{c+w}, P_{d+w}))$$

for arbitrary

$$v, w \in \mathbb{N}.$$

For example, all of the following produce the same signed output:

$$\text{Add}((P_a, P_b), (P_c, P_d)),$$

$$\text{Add}((P_{a+1}, P_{b+1}), (P_{c+1}, P_{d+1})),$$

$$\text{Add}((P_{a+5}, P_{b+5}), (P_{c+9}, P_{d+9})),$$

and

$$\text{Add}((P_{a+v}, P_{b+v}), (P_{c+w}, P_{d+w}))$$

for arbitrary v and w .

Consequently, the mapping from such input representations to the reduced signed output is not injective.

That is, if only the final reduced output

$$P_{\text{out}} = \text{Add}((P_{a+v}, P_{b+v}), (P_{c+w}, P_{d+w}))$$

is known, the output alone does not uniquely determine the values of

$$v \quad \text{and} \quad w.$$

Nor does it uniquely determine the particular unreduced input pairs used in the computation.

There are infinitely many distinct admissible input representations that may correspond to the same signed output.

This is a many-to-one property of the representation system.

It should not by itself be interpreted as a proof of computational hardness or as a statement concerning the complexity-theoretic question P versus NP . Rather, it establishes that information about the particular input representation is lost when the inputs are reduced to a single signed result.

Non-Unique Input Representations for Signed Subtraction

The same construction applies to signed subtraction.

Recall that

$$\text{Subtract}((P_a, P_b), (P_c, P_d))$$

is defined by adding the additive inverse of the second pair.

Let

$$P_v, P_w \in \mathcal{P}.$$

Then

$$(P_a, P_b)$$

and

$$(P_{a+v}, P_{b+v})$$

represent the same signed integer, while

$$(P_c, P_d)$$

and

$$(P_{c+w}, P_{d+w})$$

also represent the same signed integer.

Therefore,

$$\boxed{\text{Subtract}((P_a, P_b), (P_c, P_d)) = \text{Subtract}((P_{a+v}, P_{b+v}), (P_{c+w}, P_{d+w}))}.$$

Equivalently,

$$\boxed{\begin{aligned} &\text{Subtract}((P_a, P_b), (P_c, P_d)) \\ &= \text{Subtract}((S^{E(P_v)}(P_a), S^{E(P_v)}(P_b)), (S^{E(P_w)}(P_c), S^{E(P_w)}(P_d))) \end{aligned}}$$

Hence signed subtraction also admits infinitely many distinct unreduced input representations that produce the same reduced signed output.

Input-to-Output Representation Expansion

A standard signed addition

$$\text{Add}((P_a, P_b), (P_c, P_d))$$

requires four canonical input sets:

$$P_a, \quad P_b, \quad P_c, \quad P_d.$$

Under the non-unique representation construction, two additional canonical sets may be chosen:

$$P_v \quad \text{and} \quad P_w.$$

These determine the independent shifts applied to the two signed input pairs.

The complete representation may therefore depend upon the six canonical sets

$$\boxed{P_a, P_b, P_c, P_d, P_v, P_w.}$$

while the function ultimately returns a single signed-integer value, represented canonically by one reduced signed pair.

Thus, when the output is regarded as one signed value, the construction has a

$$\boxed{6 : 1}$$

input-set-to-output-value relationship:

$$\boxed{6 \text{ canonical input sets} \longrightarrow 1 \text{ signed output value.}}$$

The sets P_v and P_w are optional in the sense that choosing

$$P_v = P_0 \quad \text{and} \quad P_w = P_0$$

returns the original unshifted representation.

Therefore the ordinary signed operation is contained as the special case

$$v = w = 0.$$

The additional sets P_v and P_w do not alter the resulting signed value. They alter only the particular unreduced representation used to obtain that value.

This distinction between the represented value and the particular representation used to encode that value will be relevant when considering later applications involving concealed or randomized representations.